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Online distributed optimization algorithm with dynamic regret analysis under unbalanced graphs☆

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ABSTRACT

We consider online distributed convex optimization problems with a sum of locally dynamic loss functions under unbalanced graphs. When optimizing the dynamic local loss function, every node tracks the time-varying global optimal solution by communicating with neighboring nodes via communication networks in a cooperative way. We propose a novel online distributed Push–Pull algorithm and present that the proposed online optimization algorithm can track the dynamic optimal solution with proper step sizes. We analyze the dynamic regret of the proposed algorithm in two cases where the global loss function is strongly convex and smooth, or is convex, smooth and Lipschitz. Our results illustrate that the dynamic regret of the proposed online optimization algorithm can be sublinear, if the path length and the gradient variance are sublinear. At last, we demonstrate the property of the online distributed optimization algorithm by two simulation examples.

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1. Introduction

Nowadays, online distributed convex optimization problems have obtained plenty of research attention in machine learning, large-scale statistical tasks, sensor networks, and so on, such as Fosson (2020). For example, with the rapid growth of the number of images in image classification tasks, online distributed algorithms can be used to learn and track the computer's capability of classifying image in a timely manner, which is an effective tool to guarantee the accuracy of image classification after the computer learn massive quantity of images (Li, Xie, & Li, 2023). Moreover, with the increasing demand of tracking moving targets, a helpful method to solve this problem is to use multiple sensors to track the target in a cooperative way (Li et al., 2023). How to track the moving targets timely and effectively via the time-varying information collected by sensors is a key issue which needs to be solved. Since tracking moving targets in nature may be interrupted at any time, online distributed algorithms can be utilized to track moving targets and re-plan routes when encountering disturbances. Furthermore, with the emphasis on the trend of big data era, online distributed optimization algorithms are useful tools to deal with large-scale statistical tasks in the area of artificial intelligence, finance, medicine, and so on.

In general, online optimization algorithms can be classified into two categories, i.e., online centralized ones and online distributed ones. Compared with the online centralized algorithms (Hazan, Agarwal, & Kale, 2007; Yi et al., 2021; Zinkevich, August, 2003), distributed ones do not rely on a fusion center (i.e., receive every agent's information using a central unit and make a global estimation of the dynamic optimal solution), but via a network topology, such that neighboring nodes can estimate the optimal solution cooperatively by communications. In this manner, even if some nodes do not work, the estimation process will not be influenced through communications. Also, when the global loss function changes with time, dynamic regret is a useful metric that can be used to reflect the performances of online optimization algorithms. Therefore, to conduct research on the dynamic regret analysis of online distributed algorithms is of great significance.

To solve the online distributed optimization problems, researchers from all over the world have conducted a series of research works. For examples, Yi, Li, Xie, and Johansson (2020) proposed an online distributed primal–dual dynamic mirror descent algorithm with a strongly connected and balanced communication network, and demonstrated that the upper bound of the static regret is sublinear for proper step sizes. Also, Hosseini, Chapman, and Mesbahi (2013) provided an online distributed dual averaging algorithm with a balanced graph, and illustrated that the upper bound of the static regret is sublinear when

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choosing a diminishing step size. Moreover, Zhang, Ravier, Zavlanos, and Tarokh (December, 2019) proposed an online gradient tracking algorithm with a balanced graph and demonstrated that dynamic regret can be sublinear, and Akbari, Gharesifard, and Linder (2015) proposed an online distributed subgradient push-sum algorithm with an unbalanced graph and showed that the upper bound of the static regret is sublinear when choosing a specific diminishing step size, but its dynamic regret cannot achieve sublinear property while choosing a constant or diminishing step size.

Furthermore, Zhang, Shi, Zhang, and Yuan (2023) proposed an online distributed Frank–Wolfe algorithm with a balanced graph, and proved that the upper bound of the dynamic regret is sublinear when the global loss function is convex, smooth and Lipschitz, where the step size is related to the loss variance. The upper bound of the dynamic regret is $O(\sqrt{T(1+H_T)}+D_T)$, where $H_T = \sum_{t=1}^{T}\max_{i\in\mathcal{V}}\max_{x\in\mathcal{X}}|f_{i,t+1}(x)-f_{i,t}(x)|$, $D_T = \sum_{t=1}^{T}\max_{i\in\mathcal{V}}\max_{x\in\mathcal{X}}\|\nabla f_{i,t+1}(x)-\nabla f_{i,t}(x)\|$. Also, Shahrampour and Jadbabaie (2017) proposed an online distributed mirror descent algorithm with a balanced graph, and proved that its dynamic regret is sublinear when the loss function is convex and Lipschitz. The upper bound of the dynamic regret is $O(\sqrt{TC_T/(1-\sigma_2(W))})$, where $\sigma_2(W)$ is the second largest singular of matrix W and $C_T = \sum_{t=1}^{T}\|x_{t+1}^\star - Ax_t^\star\|$. Here, matrix A is known in the network, and $\{x_t^\star\}_{t=1}^T$ is the minimizer sequence. Moreover, Yang and Chen (2020) proposed an online distributed projection subgradient algorithm under an unbalanced graph and proved that the upper bound of the dynamic regret is $O(\sqrt{S_T}T^{\frac{1+a}{2}})$ with adequate step sizes when the loss function is strongly convex and the constraint set is closed, convex and bounded, where $S_T = \sum_{t=1}^{T-1}\|x_{t+1}^\star - x_t^\star\|$, and $a\in(0.5,1)$.

To sum up, most of the existing literature on the online distributed optimization problems are constrained by the following aspects. (i) Some works only consider the static regret analysis of the proposed algorithms, which cannot reflect the algorithms' capability of tracking global dynamic optimal solutions. (ii) The proposed online distributed algorithms lack the ability to learn continually, which means that their dynamic regret cannot be sublinear. (iii) Most works focus on the case where the communication network is balanced.

In order to overcome these constraints, this paper considers an online distributed optimization problem under an unbalanced graph. Specifically, the proposed algorithm first conducts a step of weighted average, and then exploits a fixed step size to assure that the algorithm can learn continually. Note that Zhang et al. (December, 2019) considered an online distributed algorithm with no constraints, where the communication network is balanced and the loss function is strongly convex and smooth, and it demonstrated that the proposed algorithm can achieve sublinear dynamic regret while step size is properly chosen. The proposed algorithm in this paper fits more practical optimization problems compared with Zhang et al. (December, 2019), Yan, Sundaram, Vishwanathan, and Qi (2012), Zhang et al. (2023), and Shahrampour and Jadbabaie (2017).

This paper refers to the convergence analysis method in Pu, Shi, Xu, and Nedić (2020), and proves that the algorithm we proposed can achieve sublinear dynamic regret. In recent years, researchers made much effort on proposing offline Push–Pull based distributed optimization algorithms. Pu (December, 2020) considered an offline robust Push–Pull algorithm, which can converge to a neighborhood of the optimal solution in expectation, and Xiong, Wu, You, and Xie (2022) proposed the Q-DGT algorithm, which can handle the situation that the communication bandwidth is limited. Since this paper considers the online convex optimization problem, higher demand on matrix analysis will be imposed on analyzing the dynamic regret of the proposed online distributed algorithm. The main contributions of this paper can be summarized as follows. (i) This paper proposes a novel online distributed optimization algorithm that can learn the optimal solution continually. The innovation of our proposed algorithm includes the following two aspects. We introduce the gradient value of the local loss function into the algorithm when time horizon are t and $t+1$ to ensure that the nodes can track the dynamic optimal solution. Besides, we introduce a scaling factor and choose adequate step sizes, and the dynamic regret of our proposed algorithm can be sublinear under comparatively weak assumptions, i.e., the loss function is convex, smooth, and Lipschitz. (ii) This paper proves that the dynamic regret of the proposed algorithm can be controlled by two terms which are relevant to the variation of the loss function under adequate assumptions, and the upper bound of the dynamic regret is $O(\sqrt{T}(1+P_T)+V_T)$ if the loss function is strongly convex and smooth, and the upper bound is $O(\sqrt{T}+P_T)$ if the loss function is convex, smooth and Lipschitz. The definitions of P_T and V_T are introduced in Section 3. (iii) Compared with those online distributed optimization algorithms under a balanced graph (Fosson, 2020; Hosseini et al., 2013; Shahrampour & Jadbabaie, 2017), the matrix theory and properties of unbalanced graphs are needed to further analyze the dynamic regret of our proposed algorithm. Specifically, "unbalanced" refers to matrices R and C, which are row stochastic and column stochastic respectively in this paper, and they are not doubly stochastic. The unbalancedness is manifested by u and v, which are the left eigenvector of the matrix R and the right eigenvector of the matrix C, respectively. We use matrix theory to eliminate the difficulties brought by u and v in the regret analysis, which increases difficulty of our analysis.

The remainder of this paper is organized as follows. In Section 2, we formulate the problem and give the assumptions needed for analyzing the dynamic regret. We propose an online distributed Push–Pull algorithm and provide results of dynamic regret analysis in Section 3, and we present the proofs of the main results in Section 4, respectively. At last, two numerical experiments are used to verify the correctness of our results in Section 5, and conclusions are presented in Section 6.

Notations: $\|\cdot\|$ denotes the 2-norm of a matrix, $\|\cdot\|_F$ denotes the F-norm of a matrix, $A^\top$ denotes the transpose of the matrix A, $\mathbb{R}^n$ denotes the n-dimensional Euclidean space, $\mathbb{R}^{m\times n}$ denotes the $m\times n$-dimensional Euclidean space, $\mathbf{I}$ denotes the n-identity matrix, $\mathbf{1}$ denotes a n-dimensional vector with all ones, and M_n denotes the set of all $n\times n$ matrix. Also, by $b_n = O(r_n)$, we mean that $\limsup_{n\to\infty}\frac{|b_n|}{r_n} < \infty$, where $\{b_n, n\geq 0\}$ is a real number sequence, and $\{r_n, n\geq 0\}$ is a positive number sequence. Moreover, $\mathcal{P}_{\mathcal{X}}(\cdot)$ denotes the projection on constraint set $\mathcal{X}$.

2. Problem formulation

Consider a communication network with n nodes, every node has the capability of communication, calculation, and storing. The online distributed optimization problem can be expressed as follows:

$$\min_{x\in\mathcal{X}} f_t(x) := \sum_{i=1}^{n} f_{i,t}(x), \qquad t=0,1,\ldots,T, \tag{1}$$

where $f_{i,t}(\cdot):\mathcal{X}\to\mathbb{R}$ is the loss function of node i at time instant t. Let $x_t^\star\in\arg\min_{x\in\mathcal{X}} f_t(x)$ be the optimal value at time instant t.

We use a tuple $\mathcal{G}=(\mathcal{V},\mathcal{E})$ to describe the network structure, where $\mathcal{V}=\{1,2,\ldots,n\}$ is the set of nodes, and $\mathcal{E}\subseteq\mathcal{V}\times\mathcal{V}$ is the set of edges. Thus, $(i,j)\in\mathcal{E}$ if and only if the ith agent can send message directly to the jth agent. We define $W=[w_{ij}]_{n\times n}\in\mathbb{R}^{n\times n}$ as a weight matrix, where $(i,j)\in\mathcal{E}$ if and only if $w_{ij}>0$. Given

a matrix $M = [m_{ij}]_{n\times n} \in \mathbb{R}^{n\times n}$, the graph induced by matrix M is noted as $\mathcal{G}_M = (\mathcal{V}, \mathcal{E}_M)$, where $\mathcal{N} = \{1, 2, \ldots, n\}$, and $(j, i) \in \mathcal{E}_M$ if and only if $m_{ij} > 0$.

For the problem (1), we need the following assumptions.

Assumption 1. $\mathcal{X}$ is the constraint set of problem (1), which is a compact, convex and bounded set. There exists a positive constant D, such that for any $x \in \mathcal{X}$, we have $\|x\| \leq D$.

Assumption 2. There exists a positive constant M, such that for any $t \in \mathbb{Z}^+$, $i \in \{1, 2, \ldots, n\}$ and $x \in \mathcal{X}$, we have $\|\nabla f_{i,t}(x)\| \leq M$.

Assumption 3. For any $t = 0, \ldots, T$, $i = 1, \ldots, n$, $f_{i,t}(x)$ is L-smooth, i.e., there is a positive constant L, such that

$$\|\nabla f_{i,t}(x) - \nabla f_{i,t}(y)\| \leq L\|x - y\|, \quad \forall x, y \in \mathcal{X}. \tag{2}$$

Assumption 4. For any $t = 0, \ldots, T$, $i = 1, \ldots, n$, $f_{i,t}(x)$ is μ-strongly convex, i.e., there is a positive constant $\mu > 0$, such that for $x, y \in \mathcal{X}$, we have

$$f_{i,t}(y) - f_{i,t}(x) \geq \nabla f_{i,t}(x)^T (y - x) + \frac{\mu}{2}\|x - y\|^2. \tag{3}$$

Assumption 5. Every $f_{i,t}(\cdot)$ is convex, i.e., for all $x, y \in \mathcal{X}$, we have

$$f_{i,t}(y) \geq f_{i,t}(x) + \nabla f_{i,t}(x)^\top (y - x).$$

Since node i can only observe $f_{i,t}(\cdot)$ at time t, nodes may communicate with each other via two communication networks $\mathcal{G}_R$ and $\mathcal{G}_C$. Every node exploits its local information and the information obtained from neighbors to estimate the optimal value $x^\star_{t+1}$ with an estimation $x_{i,t+1}$. Based on it, this process will cause the corresponding cost $f_{i,t+1}(x_{i,t+1})$. We expect to design a distributed online optimization algorithm to enable every node's state variable $x_{i,t}$ can track the dynamic optimal value $x^\star_t$. We evaluate the property of an online optimization algorithm mainly through the static regret and the dynamic regret, then we give their definitions as follows:

Definition 1. Let $\{x_{i,t}\}_{t=0}^T$ be the state variable of a given online optimization algorithm, then its static regret is $Reg_T^s \triangleq \sum_{i=1}^N \sum_{t=0}^T \left(f_t\left(x_{i,t}\right) - f_t\left(x^\star\right)\right)$, where $x^\star = \arg\min_{x\in\mathcal{X}} \sum_{t=0}^T f_t(x) \in \mathcal{X}$.

Definition 2. Let $\{x_{i,t}\}_{t=0}^T$ be the state variable of a given online optimization algorithm, then its dynamic regret is $Reg_T^d \triangleq \sum_{i=1}^N \sum_{t=0}^T \left(f_t\left(x_{i,t}\right) - f_t\left(x_t^\star\right)\right)$, where $x_t^\star = \arg\min_{x\in\mathcal{X}} f_t(x) \in \mathcal{X}$.

3. Online distributed push–pull algorithm and its convergence results

The online distributed Push–Pull algorithm is given as follows and summarized in Algorithm 1:

$$\begin{aligned} y_{i,t} &= \sum_{j=1}^n c_{ij} y_{j,t-1} + \nabla f_{i,t}(x_{i,t}) - \nabla f_{i,t-1}(x_{i,t-1}) \\ &\in \mathbb{R}^p, \end{aligned} \tag{4}$$

$$x_{i,t+1} = \mathcal{P}_{\mathcal{X}}\left[a\sum_{j=1}^n r_{ij}(x_{j,t} - \alpha y_{j,t})\right] \in \mathbb{R}^p, \tag{5}$$

where $0 < a \leq 1$ is a scaling factor, α is the step-size, with $\alpha > 0$.

Algorithm 1 Online Push-Pull

Initialization: Arbitrarily initialize $x_{i,0} \in \mathcal{X}$ and let $y_{i,0} = \nabla f_{i,0}(x_{i,0})$ for each $i \in \mathcal{V}$.

1: **for** $t = 0, 1, \cdots$ **do**
2: Update $y_{i,t}$ and $x_{i,t+1}$ via (4) and (5).
3: **end for**
4: **return** $\{x_{i,t}\}$.

Remark 1. This paper and Pu et al. (2020) studied different kinds of optimization problems. In Pu et al. (2020), researchers studied an offline distributed optimization problem:

$$\min_{x\in\mathbb{R}^p} f(x) := \sum_{i=1}^n f_i(x).$$

However, this paper studies an online distributed optimization problem:

$$\min_{x\in\mathcal{X}} f_t(x) := \sum_{i=1}^n f_{i,t}(x),\ t = 0, 1, \ldots, T.$$

The main difference between these two problems is that the local loss functions in an offline distributed optimization problem is time-invariant, but an online optimization problem considers time-variant local loss functions. Our proposed algorithm has two differences compared with Pu et al. (2020): (1) The auxiliary variable $y_{i,t}$ in our proposed algorithm involves information about the gradient of time-varying loss functions, which is $\nabla f_{i,t}(x_{i,t})$, while in that of Pu et al. (2020), it is $\nabla f_i(x_{i,t})$. (2) This paper introduces a scaling factor a, where $0 < a \leq 1$, so that our proposed algorithm can be utilized to solve optimization problems with convex, Lipschitz and smooth time-varying loss functions, and not limited to strong convex and smooth functions.

$\nabla F_t(x)$ is defined by

$$\nabla F_t(x) = [\nabla f_{1,t}(x_1), \nabla f_{2,t}(x_2), \ldots, \nabla f_{n,t}(x_n)]^\top \in \mathbb{R}^{n\times p},$$

where $x = [x_1, x_2, \ldots, x_n]^\top \in \mathbb{R}^{n\times p}$. The dynamic regret is not only related to the running time of an online optimization algorithm, but also related to the variance of the loss functions, it can be reflected by the path length and gradient variance, which are denoted by $P_T = \sum_{t=0}^{T-1} \|x^\star_{t+1} - a x^\star_t\|$, and $V_T = \sum_{t=0}^{T-1} \sup_x \|\nabla F_{t+1}(x) - \nabla F_t(x)\|$, respectively. Here, $0 < a \leq 1$.

Remark 2. Generally speaking, P_T and V_T may not have a certain correlation, since the difference between them may be large for different loss functions. For example, for some dynamic quadratic functions, when P_T is small, V_T can be large, then V_T is dominated in the result, which should be underlined.

For the analysis of the above algorithm, we also need the following assumptions on the matrices R and C.

Assumption 6. Given $R = [r_{ij}]_{n\times n} \in \mathbb{R}^{n\times n}$, $C = [c_{ij}]_{n\times n} \in \mathbb{R}^{n\times n}$, $\mathcal{G}_R$ and $\mathcal{G}_C$ are induced by R and C, and the communication topologies $\mathcal{G}_R$ and $\mathcal{G}_C$ are strongly connected, respectively. Besides, $[r_{ij}]_{n\times n}$ is a row-stochastic matrix and $[c_{ij}]_{n\times n}$ is a column-stochastic matrix, i.e., $\sum_{j=1}^n r_{ij} = 1$, $\sum_{i=1}^n c_{ij} = 1$, $\forall i \in \mathcal{V}$.

Remark 3. Since our proposed algorithm has special communication principles, we need to introduce two graphs to express the communication between agents. Thus, we use two matrices R and C to introduce corresponding graphs. The critical principle is given as follows: node i in the graph $\mathcal{G}_R$ receives its in-neighbor j's information $x_{j,t} - \alpha y_{j,t}$ at time t, i.e., the so-called "pull" process. Then, node i sends its local information $y_{i,t}$ at time t to its out-neighbor ℓ in the graph $\mathcal{G}_C$, and node ℓ receives the information, which is the so-called "push" process. Note that the graph $\mathcal{G}_R$ and the graph $\mathcal{G}_C$ can be the same.

Lemma 1 (*Xie, You, Tempo, Song, & Wu, 2018*)**.** *Under Assumption 6, there exist two vectors $u, v \in \mathbb{R}^d$, for matrices R and C such that $u^\top R = u^\top$, $Cv = v$, $u^\top \mathbf{1} = v^\top \mathbf{1} = n$, $u_i, v_i > 0$.*

Lemma 2 (Xiong et al., 2022). *Under Assumption 6, $\|R^t-\mathbf{1}u^\top/n\| \le \Lambda\gamma^t$ and $\|C^t - v^\top\mathbf{1}/n\| \le \Lambda\gamma^t$. Here, $\Lambda > 0$ and $\gamma \in (0, 1)$.*

Before presenting the convergence results, we give the aggregated form of the proposed algorithm:

$$\mathbf{y}_t = C\mathbf{y}_{t-1} + \nabla F_t(\mathbf{x}_t) - \nabla F_{t-1}(\mathbf{x}_{t-1}), \tag{6}$$

$$\mathbf{x}_{t+1} = \mathcal{P}_{\mathcal{X}^n}\Big[aR(\mathbf{x}_t - \alpha\mathbf{y}_t)\Big], \tag{7}$$

where

$$\mathbf{x}_t = [x_{1,t}, x_{2,t}, \dots, x_{n,t}]^\top \in \mathbb{R}^{n\times p},$$
$$\mathbf{y}_t = [y_{1,t}, y_{2,t}, \dots, y_{n,t}]^\top \in \mathbb{R}^{n\times p},$$
$$\mathcal{P}_{\mathcal{X}^n}(z_t) = [\mathcal{P}_{\mathcal{X}}(z_{1,t}), \dots, \mathcal{P}_{\mathcal{X}}(z_{n,t})]^\top \in \mathbb{R}^{n\times p}, \tag{8}$$

for any $z_t = [z_{1,t}, \dots, z_{n,t}]^\top \in \mathbb{R}^{n\times p}$.

In the following part, we give the main convergence results of this paper.

Theorem 1. *For the optimization problem (1), and the online distributed optimization algorithm (4) and (5) with $a = 1$ and $\alpha = \frac{c}{\sqrt{T}}$, where c is a positive constant, suppose that Assumptions 1, 2, 3, 4, and 6 hold, we have*

$$\begin{aligned}
&Reg_T^d \\
&\le \frac{1}{u^\top v}\Bigg[\sqrt{T}\frac{\sqrt{n}\|u\|\|\mathbf{x}_0 - \mathbf{1}(x_0^\star)^\top\|}{2c} \\
&+ \|u\|\sqrt{n}\frac{\Lambda DnM(2\Lambda+1-\gamma)}{(1-\gamma)^2} + 2D\sqrt{n}nM\frac{\Lambda(2\Lambda+1-\gamma)}{(1-\gamma)^2} \\
&+ 4D^2L\sqrt{n}\Lambda\frac{1}{(1-\gamma)^2} + nDL\frac{\Lambda D}{1-\gamma} \\
&+ \Bigg(\Big(\frac{1}{2} + \|u\|\sqrt{n}\Lambda\Big)\Big(nM\frac{2\Lambda+1-\gamma}{1-\gamma}\Big)^2 \\
&+ \Big(\frac{4DL\sqrt{n}\Lambda^2}{(1-\gamma)^2} + 2u^\top vDLn\Lambda\Big)nM\frac{2\Lambda+1-\gamma}{1-\gamma} \\
&+ u^\top v\Lambda n^2DLM\frac{2\Lambda+1-\gamma}{1-\gamma}\Bigg)c\sqrt{T} \\
&+ \Bigg(\frac{2nD\sqrt{T}}{c} + n^2M\frac{2\Lambda+1-\gamma}{1-\gamma}\Bigg)P_T + \frac{\sqrt{n}\Lambda}{1-\gamma}V_T\Bigg].
\end{aligned} \tag{9}$$

Remark 4. By the upper bound of dynamic regret in Theorem 1, we can adjust the step size to reduce the running time of the proposed algorithm according to the required accumulative error, the information about the loss functions and optimal solutions.

Remark 5. From (9) in the proof of Theorem 1, we can see that when the loss function is strongly convex and smooth, both P_T and V_T have an effect on the upper bound of the dynamic regret. If the global loss function $f_t(\cdot)$ varies slowly, and P_T, V_T have small upper bounds. Note that if P_T is smaller than $O(\sqrt{T})$ and V_T is smaller than $O(T)$, then the bound is sublinear. However, if the optimal solution $x_t^\star$ fluctuates drastically, P_T or V_T may be bigger than $O(T)$, then the proposed algorithm cannot ensure a sublinear bound. In other words, under some extreme cases, online optimization problems cannot be solved. When our assumptions on the loss function are the same as Yang and Chen (2020), our upper bound is better than Yang and Chen (2020) when $V_T = O(P_T)$(e.g., if the loss function is some class of time-varying quadratic convex function).

Theorem 2. *For the optimization problem (1), and the online distributed optimization algorithm (4) and (5), suppose that Assumptions 1, 2, 5 and 6 hold, $0 < a < 1$, $\alpha = \frac{c}{\sqrt{T}}$, where c is a positive constant, we have*

$$\begin{aligned}
&Reg_T^d \\
&\le nM\sum_{t=0}^{T}\Bigg(n\Lambda D/(1-a\gamma) \\
&+ \Lambda\alpha n^2MT\frac{2\Lambda+1-\gamma}{(1-\gamma)(1-a\gamma)} \\
&+ \frac{1}{(1-a)n}\Bigg(\sum_{t=0}^{T-1}\|x_{t+1}^\star - ax_t^\star\| \\
&+ \alpha nMT\frac{2\Lambda+1-\gamma}{1-\gamma} + \sqrt{n}\|u\|\|\mathbf{x}_0 - \mathbf{1}(x_0^\star)^\top\|\Bigg)\Bigg).
\end{aligned} \tag{10}$$

Remark 6. From (10), we can see that when the loss function is convex, smooth and Lipschitz, both P_T and V_T have a big impact on the upper bound of the dynamic regret. When our assumptions on the loss function are the same as Zhang et al. (2023), our upper bound is better than Zhang et al. (2023) when $P_T = O(D_T)$ (e.g., if the loss function is a time-varying quadratic convex function); When our assumptions on the loss function are similar to Shahrampour and Jadbabaie (2017), our upper bound is better than Shahrampour and Jadbabaie (2017) when $P_T = O(\sqrt{T})$. (e.g., if the loss function is some kind of time-varying quadratic convex function)

Remark 7. For examples, the online distributed Frank Wolfe algorithm proposed by Zhang et al. (2023) uses the gradient tracking technology to track the variable $\nabla F_t(\hat{x}_{i,t})$, instead of $\nabla F_t(x_{i,t})$. Then, the centralized Frank Wolfe algorithm is proposed to track the time-varying global optimal solution. According to the operation mechanism of the centralized Frank Wolfe algorithm, the decision variable $\hat{x}_{i,t}$ can track time-varying optimal solutions well. Although $x_{i,t}$ can also track time-varying optimal solutions, the dynamic regret will be correspondingly larger. Also, the auxiliary variable $y_{i,t}$ in our paper precisely estimates $\nabla f_t(x_{i,t})$, so $x_{i,t}$ can track the time-varying optimal solution more accurately. The online distributed mirror descent algorithm proposed by Shahrampour and Jadbabaie (2017) involves the Bregman divergence, resulting in related errors and an increase in the dynamic regret. In addition, according to the analysis of linear minimization steps, each $\hat{x}_{i,t}$ tracks the local time-varying optimal solution $x_{i,t}^* \in \arg\min_{x\in\mathcal{X}} f_{i,t}(x)$, which also causes errors and leads to an increase in the dynamic regret. Thus, the performance of our proposed algorithm can be better than these proposed algorithms.

4. Proofs of the main results

In this section, we study the properties of dynamic regret of our proposed algorithm. Define

$$\bar{x}_t := \frac{1}{n}u^\top\mathbf{x}_t \in \mathbb{R}^{1\times p},$$
$$\bar{y}_t := \frac{1}{n}\mathbf{1}^\top\mathbf{y}_t \in \mathbb{R}^{1\times p},$$
$$g_t := \frac{1}{n}\mathbf{1}^\top\nabla F_t(\mathbf{1}\bar{x}_t) \in \mathbb{R}^{1\times p},$$
$$\xi_t := \mathcal{P}_{\mathcal{X}^n}\Big[aR(\mathbf{x}_t - \alpha\mathbf{y}_t)\Big] - aR(\mathbf{x}_t - \alpha\mathbf{y}_t),$$

$\epsilon_t := \xi_t - aR\alpha \mathbf{y}_t$.

4.1. Proof of Theorem 1

Lemma 3. *For the optimization problem (1), and the online distributed optimization algorithm (4) and (5), when $a = 1$, suppose that Assumptions 2 and 6 hold, we have*

$$\|\xi_t\| \le \alpha nM\frac{2\Lambda+1-\gamma}{1-\gamma}, \tag{11}$$

$$\|\mathbf{x}_t - \mathbf{1}\bar{x}_t\| \le \Lambda D\gamma^{t-1} + \Lambda\alpha nM\frac{2\Lambda+1-\gamma}{(1-\gamma)^2}, \tag{12}$$

and

$$\|\mathbf{x}_t - \mathbf{x}_{t-1}\| \le 2\Lambda D\gamma^{t-2} + 2\Lambda\alpha nM\frac{2\Lambda+1-\gamma}{(1-\gamma)^2}, \tag{13}$$

where α is the step size of our proposed algorithm, Λ and γ are the numbers defined in Lemma 2.

Proof. Defining $s_{t+1} = \nabla F_{t+1}(\mathbf{x}_{t+1}) - \nabla F_t(\mathbf{x}_t)$, we rewrite (6) as follows:

$$\mathbf{y}_{t+1} = C\mathbf{y}_t + s_{t+1}.$$

Then, we have

$$\mathbf{y}_{t+1} = C^{t+1}\mathbf{y}_0 + \sum_{l=1}^{t+1} C^{t+1-l}s_l. \tag{14}$$

From $y_{i,0} = \nabla f_{i,0}(x_{i,0})$ and $s_l = \nabla F_l(\mathbf{x}_l) - \nabla F_{l-1}(\mathbf{x}_{l-1})$, we know that

$$\begin{aligned}\|y_{t+1}\| \le& \sum_{l=0}^{t}\|C^{t-l} - C^{t-l+1}\|\|\nabla F_l(\mathbf{x}_l)\| \\ &+ \|\nabla F_{t+1}(\mathbf{x}_{t+1})\|,\end{aligned} \tag{15}$$

where $C^0 := \mathbf{I}$. According to Assumption 2, we have $\|\nabla F_t(\mathbf{x}_t)\| \le \sqrt{n}M$. Then, by Lemma 2, we know that $\|y_{t+1}\| \le \sqrt{n}M\frac{2\Lambda+1-\gamma}{1-\gamma}$. From the convexity of $\mathcal{X}$ and Assumption 6, we know that $\sum_{j=1} r_{ij}x_{j,t} \in \mathcal{X}$. Therefore, $\|\xi_t\| \le \sqrt{n}\alpha\|y_t\|$ and we can get (11) from (15). From (7), we have

$$\begin{aligned}\mathbf{x}_{t+1} &= R\mathbf{x}_t + \epsilon_t \\ &= R^{t+1}\mathbf{x}_0 + \sum_{l=1}^{t} R^{l-1}\epsilon_{t+1-l}.\end{aligned} \tag{16}$$

Multiplying $\mathbf{1}u^\top/n$ on both sides of (7), we have

$$\mathbf{1}\bar{x}_t = \frac{1}{n}\mathbf{1}u^\top\mathbf{x}_1 + \frac{1}{n}\sum_{l=1}^{t}\mathbf{1}u^\top\epsilon_{t+1-l}. \tag{17}$$

By (16) and (17), we know that

$$\begin{aligned}&\|\mathbf{x}_t - \mathbf{1}\bar{x}_t\| \\ \le&\|R^t - \mathbf{1}u^\top/n\|\|\mathbf{x}_1\| + \Lambda\sum_{l=1}^{t}\|R^{l-1} - \mathbf{1}u^\top/n\|\|\epsilon_{t+1-l}\| \\ \le&\Lambda D\gamma^t + \Lambda\sum_{l=1}^{t}\gamma^{l-1}\|\epsilon_{t+1-l}\| \\ \le&\Lambda D\gamma^t + \Lambda\alpha nM\frac{2\Lambda+1-\gamma}{(1-\gamma)^2},\end{aligned}$$

where the last inequality follows from Lemmas 1 and 2, then we obtain (12). Besides, when we let t to be $t-1$ in (16) and using Lemma 2, it yields (13).

Lemma 4. *For the optimization problem (1), and the online distributed optimization algorithm (4) and (5), when $a = 1$, suppose that Assumptions 2, 3 and 6 hold, we have*

$$\begin{aligned}&\|u^\top(v\mathbf{1}^\top\nabla F_t(\mathbf{x}_t)/n - \mathbf{y}_t)\| \\ \le&\sqrt{n}nM\frac{2\Lambda+1-\gamma}{1-\gamma}\Lambda\gamma^t \\ &+ \sqrt{n}\Lambda\sum_{l=1}^{t+1}\gamma^{t+1-l}\sup_x\|\nabla F_l(x) - \nabla F_{l-1}(x)\| \\ &+ 2\sqrt{n}\Lambda^2LD(t+1)\gamma^{t-1} \\ &+ 2\Lambda^2\alpha nM\frac{2\Lambda+1-\gamma}{(1-\gamma)^3},\end{aligned}$$

where u and v are defined in Lemma 1.

Proof. For (6), we have

$$\begin{aligned}v\mathbf{1}^\top[\mathbf{y}_t - \nabla F_t(\mathbf{x}_t)] &= v\mathbf{1}^\top[\mathbf{y}_{t-1} - \nabla F_{t-1}(\mathbf{x}_{t-1})] \\ &= v\mathbf{1}^\top[\mathbf{y}_0 - \nabla F_0(\mathbf{x}_0)] = 0.\end{aligned}$$

Therefore,

$$v\mathbf{1}^\top\nabla F_t(\mathbf{x}_t) - \mathbf{y}_t = v\mathbf{1}^\top\mathbf{y}_t - \mathbf{y}_t. \tag{18}$$

According to (12), we have

$$v\mathbf{1}^\top y_t = v\mathbf{1}^\top\mathbf{y}_0 - \sum_{l=1}^{t} v\mathbf{1}^\top s_l. \tag{19}$$

Left multiplying $u^\top$ on the both sides of (18), and according to (14) and (19), we have

$$\begin{aligned}&u^\top(v\mathbf{1}^\top\nabla F_t(\mathbf{x}_t)/n - \mathbf{y}_t) \\ =& -u^\top(C^t - v\mathbf{1}^\top/n)\mathbf{y}_0 + \sum_{l=1}^{t+1}u^\top(C^{t+1-l} - v\mathbf{1}^\top/n)s_l.\end{aligned}$$

Thus, we have

$$\begin{aligned}&\|u^\top(v\mathbf{1}^\top\nabla F_t(\mathbf{x}_t)/n - \mathbf{y}_t)\| \\ \le&\sqrt{n}nM\frac{2\Lambda+1-\gamma}{1-\gamma}\|C^t - v\mathbf{1}^\top/n\| \\ &+ \sqrt{n}\sum_{l=1}^{t+1}\|C^{t+1-l} - v\mathbf{1}^\top/n\|\Big(L\|\mathbf{x}_l - \mathbf{x}_{l-1}\| \\ &+ \sup_x\|\nabla F_l(x) - \nabla F_{l-1}(x)\|\Big).\end{aligned}$$

By Lemmas 2 and 3, we can get the conclusion.

Lemma 5. *For the optimization problem (1), and the online distributed optimization algorithm (4) and (5), when $a = 1$, suppose that Assumptions 1, 2 and 6 hold, we have*

$$\begin{aligned}&\sum_{i=1}^{n}u_i\|x_{i,t+1} - x^\star_{t+1}\|^2 \\ \le&\sum_{i=1}^{n}u_i\|x_{i,t} - x^\star_t\|^2 + 4nD\|x^\star_{t+1} - x^\star_t\| \\ &+ \alpha^2(nM\frac{2\Lambda+1-\gamma}{1-\gamma})^2 - 2\alpha\Big[\sum_{i=1}^{n}u_i(\sum_{j=1}^{n}r_{ij}x_{j,t} - \bar{x}_t, y_{i,t}) \\ &+ (\bar{x}_t - x^\star_t, \sum_{i=1}^{n}u_iy_{i,t} - u^\top v\sum_{i=1}^{n}\nabla f_{i,t}(x_{i,t}))\end{aligned}$$

$$+ u^\top v(\bar{x}_t - x_t^\star, \sum_{i=1}^n(\nabla f_{i,t}(x_{i,t}) - \nabla f_{i,t}(\bar{x}_t)))$$
$$+ u^\top v(\bar{x}_t - x_t^\star, \sum_{i=1}^n \nabla f_{i,t}(\bar{x}_t)) + 6nD\|x_{t+1}^\star - x_t^\star\|\Big]$$
$$+ 2\alpha n^2\sqrt{n}M\frac{2\Lambda+1-\gamma}{1-\gamma}\|x_{t+1}^\star - x_t^\star\|,$$

where u_i is the ith component of vector u.

Proof. By (5), Assumption 1, the fact $x_{i,t} \in \mathcal{X}, x_t^\star \in \mathcal{X}$ and Lemma 3, we have

$$\sum_{i=1}^n u_i\|x_{i,t+1} - x_{t+1}^\star\|^2$$
$$\le \sum_{i=1}^n u_i\|\sum_{j=1}^n r_{ij}x_{j,t} - x_{t+1}^\star - \alpha y_{i,t}\|^2$$
$$\le \sum_{i=1}^n u_i\|x_{i,t} - x_{t+1}^\star\|^2 + \alpha^2(nM\frac{2\Lambda+1-\gamma}{1-\gamma})^2$$
$$- 2\alpha\sum_{i=1}^n u_i\Big((\sum_{j=1}^n r_{ij}x_{j,t} - x_t^\star, y_{i,t}) + (x_t^\star - x_{t+1}^\star, y_{i,t})\Big)$$
$$\le \sum_{i=1}^n u_i\|x_{i,t} - x_t^\star\|^2 + 4nD\|x_{t+1}^\star - x_t^\star\|$$
$$+ \alpha^2(nM\frac{2\Lambda+1-\gamma}{1-\gamma})^2 - 2\alpha\Big[\sum_{i=1}^n u_i(\sum_{j=1}^n r_{ij}x_{j,t} - \bar{x}_t, y_{i,t})$$
$$+ (\bar{x}_t - x_t^\star, \sum_{i=1}^n u_i y_{i,t} - u^\top v\sum_{i=1}^n \nabla f_{i,t}(x_{i,t}))$$
$$+ u^\top v(\bar{x}_t - x_t^\star, \sum_{i=1}^n(\nabla f_{i,t}(x_{i,t}) - \nabla f_{i,t}(\bar{x}_t)))$$
$$+ u^\top v(\bar{x}_t - x_t^\star, \sum_{i=1}^n \nabla f_{i,t}(\bar{x}_t)) + 6nD\|x_{t+1}^\star - x_t^\star\|\Big]$$
$$+ 2\alpha n^2\sqrt{n}M\frac{2\Lambda+1-\gamma}{1-\gamma}\|x_{t+1}^\star - x_t^\star\|.$$

Lemma 6. *For the optimization problem (1), and the online distributed optimization algorithm (4) and (5), when $a = 1$, suppose that Assumptions 1, 2, 4 and 6 hold, we have*

$$\sum_{t=0}^T[f_t(\bar{x}_t) - f_t(x_t^\star)]$$
$$\le \frac{1}{u^\top v}\Bigg[\frac{\sqrt{n}\|u\|\|\mathbf{x}_0 - \mathbf{1}(x_0^\star)^\top\|}{2\alpha}$$
$$+ \|u\|\sqrt{n}\frac{\Lambda DnM(2\Lambda+1-\gamma)}{(1-\gamma)^2} + 2D\sqrt{n}nM\frac{\Lambda(2\Lambda+1-\gamma)}{(1-\gamma)^2}$$
$$+ 4D^2L\sqrt{n}\Lambda\frac{1}{(1-\gamma)^2}$$
$$+ \Big((\frac{1}{2} + \|u\|\sqrt{n}\Lambda)(nM\frac{2\Lambda+1-\gamma}{1-\gamma})^2$$
$$+ (\frac{4DL\sqrt{n}\Lambda^2}{(1-\gamma)^2} + 2u^\top vDLn\Lambda)nM\frac{2\Lambda+1-\gamma}{1-\gamma}\Big)\alpha T$$
$$+ \Big(\frac{2nD}{\alpha} + n^2M\frac{2\Lambda+1-\gamma}{1-\gamma}\Big)P_T + \frac{\sqrt{n}\Lambda}{1-\gamma}V_T. \quad (20)$$

Proof. Recalling Assumption 4, we have

$$(\bar{x}_t - x_t^\star, \nabla f_{i,t}(\bar{x}_t))$$
$$\ge f_{i,t}(\bar{x}_t) - f_{i,t}(x_t^\star) + \frac{\mu}{2}\|\bar{x}_t - x_t^\star\|^2. \quad (21)$$

By Lemma 5 and (21), we know that

$$u^\top v\sum_{i=1}^n[f_{i,t}(\bar{x}_t) - f_{i,t}(x_t^\star)]$$
$$\le \frac{1}{2\alpha}\sum_{i=1}^n u_i[\|x_{i,t} - x_t^\star\|^2 - \|x_{i,t+1} - x_{t+1}^\star\|^2]$$
$$+ \frac{\alpha}{2}(nM\frac{2\Lambda+1-\gamma}{1-\gamma})^2 - \sum_{i=1}^n u_i(\sum_{j=1}^n r_{ij}x_{j,t} - \bar{x}_t, y_{i,t})$$
$$- (\bar{x}_t - x_t^\star, \sum_{i=1}^n u_i y_{i,t} - u^\top v\sum_{i=1}^n \nabla f_{i,t}(x_{i,t}))$$
$$- u^\top v(\bar{x}_t - x_t^\star, \sum_{i=1}^n(\nabla f_{i,t}(x_{i,t}) - \nabla f_{i,t}(\bar{x}_t)))$$
$$+ \frac{2nD\|x_{t+1}^\star - x_t^\star\|}{\alpha} + n^2M\frac{2\Lambda+1-\gamma}{1-\gamma}\|x_t^\star - x_{t+1}^\star\|$$
$$+ \sqrt{n}\Lambda\sum_{l=1}^{t+1}\gamma^{t+1-l}\sup_x\|\nabla F_l(x) - \nabla F_{l-1}(x)\|. \quad (22)$$

Taking the summation from $t = 1$ to T, from Assumption 1, Lemmas 3 and 4, we get

$$\sum_{t=0}^T[f_t(\bar{x}_t) - f_t(x_t^\star)]$$
$$\le \frac{1}{u^\top v}\Bigg[\frac{\sqrt{n}\|u\|\|\mathbf{x}_0 - \mathbf{1}(x_0^\star)^\top\|}{2\alpha}$$
$$+ \|u\|\sqrt{n}\frac{\Lambda DnM(2\Lambda+1-\gamma)}{(1-\gamma)^2} + 2D\sqrt{n}nM\frac{\Lambda(2\Lambda+1-\gamma)}{(1-\gamma)^2}$$
$$+ 4D^2L\sqrt{n}\Lambda\frac{1}{(1-\gamma)^2}$$
$$+ \Big((\frac{1}{2} + \|u\|\sqrt{n}\Lambda)(nM\frac{2\Lambda+1-\gamma}{1-\gamma})^2$$
$$+ (\frac{4DL\sqrt{n}\Lambda^2}{(1-\gamma)^2} + 2u^\top vDLn\Lambda)nM\frac{2\Lambda+1-\gamma}{1-\gamma}\Big)\alpha T$$
$$+ \Big(\frac{2nD}{\alpha} + n^2M\frac{2\Lambda+1-\gamma}{1-\gamma}\Big)P_T + \frac{\sqrt{n}\Lambda}{1-\gamma}V_T. \quad (23)$$

Here, we give the proof of Theorem 1, where we use the conclusion of Lemmas 3 and 6.

Proof. From Assumptions 1, 3, Lemmas 3, 6, and Definition 2, we know that

$$Reg_T^d = \sum_{t=0}^T\sum_{i=1}^n f_t(x_{i,t}) - f_t(x_t^\star)$$
$$\le \frac{L}{2}\sum_{t=0}^T\sum_{i=1}^n\|x_{i,t} - \bar{x}_t\|^2 + \sum_{t=0}^T[f_t(\bar{x}_t) - f_t(x_t^\star)]$$
$$\le \frac{1}{u^\top v}\Bigg[\frac{\sqrt{n}\|u\|\|\mathbf{x}_0 - \mathbf{1}(x_0^\star)^\top\|}{2\alpha}$$
$$+ \|u\|\sqrt{n}\frac{\Lambda DnM(2\Lambda+1-\gamma)}{(1-\gamma)^2}$$
$$+ 2D\sqrt{n}nM\frac{\Lambda(2\Lambda+1-\gamma)}{(1-\gamma)^2}$$

$$+ 4D^2L\sqrt{n}\Lambda\frac{1}{(1-\gamma)^2} + nDL\frac{\Lambda D}{1-\gamma}$$
$$+ \Big((\frac{1}{2} + \|u\|\sqrt{n}\Lambda)(nM\frac{2\Lambda+1-\gamma}{1-\gamma})^2$$
$$+ (\frac{4DL\sqrt{n}\Lambda^2}{(1-\gamma)^2} + 2u^\top vDLn\Lambda)nM\frac{2\Lambda+1-\gamma}{1-\gamma}$$
$$+ u^\top v\Lambda n^2DLM\frac{2\Lambda+1-\gamma}{1-\gamma}\Big)\alpha T$$
$$+ \Big(\frac{2nD}{\alpha} + n^2M\frac{2\Lambda+1-\gamma}{1-\gamma}\Big)P_T + \frac{\sqrt{n}\Lambda}{1-\gamma}V_T\Big]. \quad (24)$$

The proof is completed.

4.2. Proof of Theorem 2

Lemma 7. *For the optimization problem (1), and the online distributed optimization algorithm (4) and (5), when $0 < a < 1$, suppose that Assumptions 1, 2 and 6 hold, we have*

$$\|\xi_t\| \le \alpha nM\frac{2\Lambda+1-\gamma}{1-\gamma}, \quad (25)$$

and

$$\|\mathbf{x}_t - \mathbf{1}\bar{x}_t\| \le \Lambda D(a\gamma)^t + \Lambda\alpha nM\frac{2\Lambda+1-\gamma}{(1-\gamma)(1-a\gamma)}. \quad (26)$$

Proof. The proof of (25) is the same as that of Lemma 3, we omit it here. From (7), we have

$$\mathbf{x}_{t+1} = aR\mathbf{x}_t + \epsilon_t$$
$$= a^{t+1}R^{t+1}\mathbf{x}_0 + \sum_{l=1}^{t} a^{l-1}R^{l-1}\epsilon_{t+1-l}. \quad (27)$$

Multiplying $\mathbf{1}u^\top/n$ on both sides of (7), we have

$$\mathbf{1}\bar{x}_t = \frac{a^{t+1}}{n}\mathbf{1}u^\top\mathbf{x}_0 + \frac{1}{n}\sum_{l=1}^{t}\mathbf{1}u^\top a^{l-1}\epsilon_{t+1-l}. \quad (28)$$

By (27) and (28), we know that

$$\|\mathbf{x}_t - \mathbf{1}\bar{x}_t\|$$
$$\le a^{t+1}\|R^t - \mathbf{1}u^\top/n\|\|\mathbf{x}_0\|$$
$$+ \Lambda\sum_{l=1}^{t} a^{l-1}\|R^{l-1} - \mathbf{1}u^\top/n\|\|\epsilon_{t+1-l}\|$$
$$\le \Lambda D(a\gamma)^t + \Lambda\sum_{l=1}^{t}(a\gamma)^{l-1}\|\epsilon_{t+1-l}\|$$
$$\le \Lambda D(a\gamma)^t + \Lambda\alpha nM\frac{2\Lambda+1-\gamma}{(1-\gamma)(1-a\gamma)},$$

where the last inequality follows from Lemmas 1 and 2, then we obtain (26).

Here, we give the proof of Theorem 2, which uses the conclusion in Lemmas 1 and 7.

Proof. According to online distributed optimization algorithm (4), (5), Lemmas 1 and 7, we have

$$\sum_{i=1}^{n} u_i\|x_{i,t+1} - x^\star_{t+1}\|$$
$$\le \sum_{i=1}^{n} u_i\|\sum_{j=1}^{n} ar_{ij}x_{j,t} - x^\star_{t+1} - \alpha y_{i,t}\|$$
$$\le \sum_{i=1}^{n} u_i\|ax_{i,t} - x^\star_{t+1}\| + \alpha nM\frac{2\Lambda+1-\gamma}{1-\gamma}$$
$$\le a\sum_{i=1}^{n} u_i\|x_{i,t} - x^\star_t\| + \|x^\star_{t+1} - ax^\star_t\|$$
$$+ \alpha nM\frac{2\Lambda+1-\gamma}{1-\gamma}. \quad (29)$$

From Assumption 5, we have

$$f_t(x_{i,t}) - f_t(x^\star_t) \le \nabla f_t(x_{i,t})^\top(x_{i,t} - x^\star_t). \quad (30)$$

By Assumption 2 and (30), we have

$$f_t(x_{i,t}) - f_t(x^\star_t) \le \|\nabla f_t(x_{i,t})\|\|x_{i,t} - x^\star_t\|$$
$$= \|\sum_{j=1}^{n}\nabla f_{j,t}(x_{i,t})\|\|x_{i,t} - x^\star_t\|$$
$$\le nM\|x_{i,t} - x^\star_t\|. \quad (31)$$

From (31), Lemma 7 and Definition 2, we have

$$Reg^d_T = \sum_{t=0}^{T}\sum_{i=1}^{n} f_t(x_{i,t}) - f_t(x^\star_t)$$
$$\le nM\sum_{t=0}^{T}\sum_{i=1}^{n}\|x_{i,t} - x^\star_t\|$$
$$\le nM\sum_{t=0}^{T}\sum_{i=1}^{n}\Big(\|x_{i,t} - \bar{x}_t\| + \|\bar{x}_t - x^\star_t\|\Big)$$
$$\le nM\sum_{t=0}^{T}\Big(\sqrt{n}\|\mathbf{x}_t - \mathbf{1}\bar{x}_t\|_F + \sum_{i=1}^{n}\frac{u_i}{n}\|x_{i,t} - x^\star_t\|\Big)$$
$$\le nM\sum_{t=0}^{T}\Big(n\|\mathbf{x}_t - \mathbf{1}\bar{x}_t\| + \sum_{i=1}^{n}\frac{u_i}{n}\|x_{i,t} - x^\star_t\|\Big). \quad (32)$$

Taking summation from $t = 0$ to $T-1$ on both sides of (29), by Assumption 1, we have

$$(1-a)\sum_{t=0}^{T}\sum_{i=1}^{n} u_i\|x_{i,t} - x^\star_t\|$$
$$\le \sum_{t=0}^{T-1}\|x^\star_{t+1} - ax^\star_t\|$$
$$+ \alpha nMT\frac{2\Lambda+1-\gamma}{1-\gamma} + \sum_{i=1}^{n} u_i\|x_{i,0} - x^\star_0\|$$
$$\le \sum_{t=0}^{T-1}\|x^\star_{t+1} - ax^\star_t\|$$
$$+ \alpha nMT\frac{2\Lambda+1-\gamma}{1-\gamma} + \|u\|\|\mathbf{x}_0 - \mathbf{1}(x^\star_t)^\top\|_F$$
$$\le \sum_{t=0}^{T-1}\|x^\star_{t+1} - ax^\star_t\|$$
$$+ \alpha nMT\frac{2\Lambda+1-\gamma}{1-\gamma} + \sqrt{n}\|u\|\|\mathbf{x}_0 - \mathbf{1}(x^\star_t)^\top\|. \quad (33)$$

By Lemma 7, (32) and (33), we know that

$$Reg^d_T \le nM\sum_{t=0}^{T}\Big(n\Lambda D/(1-a\gamma)$$
$$+ \Lambda\alpha n^2MT\frac{2\Lambda+1-\gamma}{(1-\gamma)(1-a\gamma)}$$

$$+\frac{1}{(1-a)n}\left(\sum_{t=0}^{T-1}\|x_{t+1}^{\star}-ax_t^{\star}\|\right.$$
$$\left.+\alpha nMT\frac{2\Lambda+1-\gamma}{1-\gamma}+\sqrt{n}\|u\|\|\mathbf{x}_0-\mathbf{1}(x_0^{\star})^{\top}\|\right).$$

The proof is completed.

5. Numerical simulation

In this section, we use two numerical experiments to verify our theoretical results.

Example 1. Consider the following online convex optimization problem for localization in sensor networks:

$$\min_{x\in\mathcal{X}}\sum_{i=1}^{n}\frac{1}{2}(P_ix-q_{i,t})^2,\quad t=1,2,\ldots,T,$$

where $\mathcal{X}=\{x\in\mathbb{R}|-1\le x\le 1\}$. Considering a communication network with n sensors, each sensor $i\in\{1,\ldots,n\}$ receives an observation scalar $q_{i,t}$ at time slot t. In this model, we assume the actual value $s=0.0095$ is not available to sensors and $q_{i,t}=a_{i,t}s+b_{i,t}$, where $a_{i,t}$ and $b_{i,t}$ are randomly generated from a uniform distribution on [0,2] and [-0.01,0.01] respectively. We also have $P_i=1$, $i\in\{1,\ldots,n\}$ which is the expectation of $a_{i,t}$. The communication topology is induced by matrix R and C, when $n=5$, $R=\begin{pmatrix}0.2&0.2&0.2&0.2&0.2\\0&0.5&0.5&0&0\\0&0&0.5&0&0.5\\0&0.5&0&0.5&0\\0.25&0&0&0.25&0.5\end{pmatrix}$.

It is easy to verify that R meets $R\mathbf{1}=\mathbf{1}$, i.e., R is row-stochastic. Take $C=R^{\top}$, it is easy to see that $\mathbf{1}^{\top}C=\mathbf{1}^{\top}$, i.e., C is column-stochastic.

The communication network topology used in this experiment is shown in Fig. 1. From Fig. 2, we know that when $\alpha=0.02$, the average dynamic regret is bigger than that of 0.04, and both of them converge to 0. Fig. 3 shows the tracking performances when $\alpha=0.04$, cooperation between sensors result in consensus, and the consensus value is 0.0095. Fig. 4 shows the average dynamic regret of proposed algorithm and its centralized counterpart, and we can see that difference between them are small. As the time horizon T is large, two curves are close to each other. When $n=6$, $R=\begin{pmatrix}0.1&0.3&0.2&0.2&0.2&0\\0&0.5&0.4&0&0&0.1\\0&0&0.5&0&0.5&0\\0&0.5&0&0.5&0&0\\0.25&0&0&0.25&0.5&0\\0&0&0&0.5&0&0.5\end{pmatrix}$, $C=R^{\top}$. From Fig. 5, we know that when n is larger, the average dynamic regret is also larger, which corresponds to the result in Theorem 1. Fig. 6 shows that proposed algorithm outperforms Yang and Chen (2020) while setting our step size to be 0.025, while the step size of Yang and Chen (2020) is set to be 0.002.

Example 2. In the second numerical experiment, we consider the following online convex optimization problem:

$$\min_{x\in\mathcal{X}}\frac{1}{2}\sum_{i=1}^{n}P_i\|x-q_{i,t}\|^2,\quad t=1,2,\ldots,T,$$

where $\mathcal{X}=\{x\in\mathbb{R}^2|\|x\|\le 2\}$. We assume the actual value $x_t^{\star}$ is not available to sensors, where $x_{t+1}^{\star}=ax_t^{\star}+v_t$, $a=0.95$, $P_1=0$, $P_2=P_3=P_4=P_5=1$. $q_{i,t}=x_t^{\star}+b_{i,t}$ and every component of $b_{i,t}$ and v_t is randomly generated from $[-0.01,0.01]$.

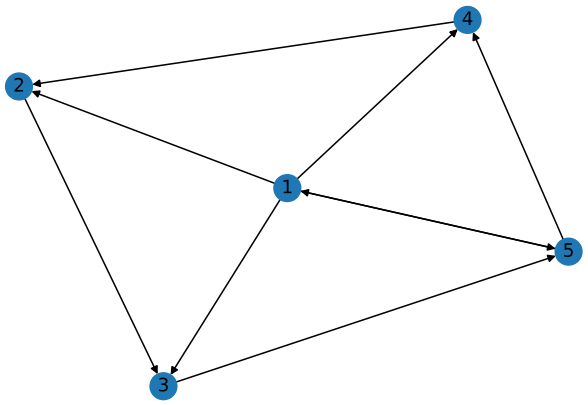


Fig. 1. The communication network topology used in two experiments when $n=5$.

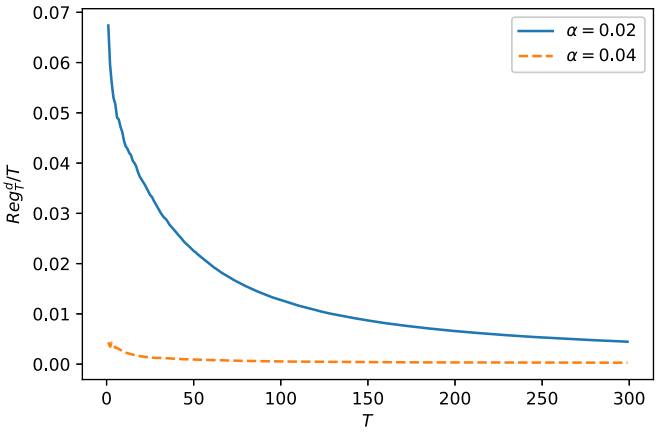


Fig. 2. Performances of proposed algorithm under different step sizes in the first experiment.

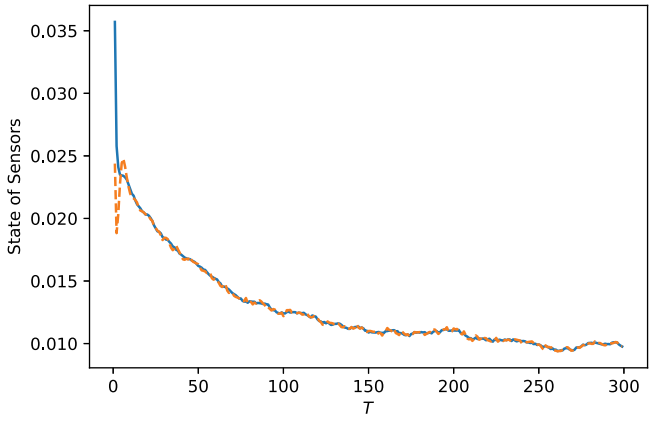


Fig. 3. Sensors' state estimation versus time for two of the sensors in the first experiment.

Fig. 7 shows the average dynamic regret of proposed algorithm, we can find that the average dynamic regret of proposed algorithm is decreasing continuously and it is close to 0 finally. Fig. 8 shows the average dynamic regret of proposed algorithm and its centralized counterpart, and we can see that difference between them are small too. As the time horizon T is large, two curves are close to each other. When $n=7$, $R=$

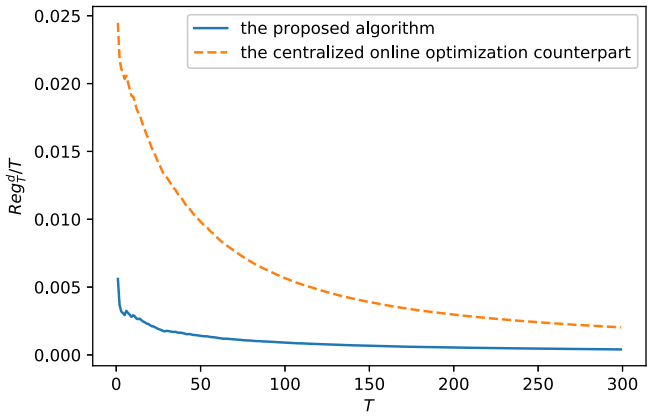


Fig. 4. Performances of proposed algorithm and its centralized counterpart in the first experiment.

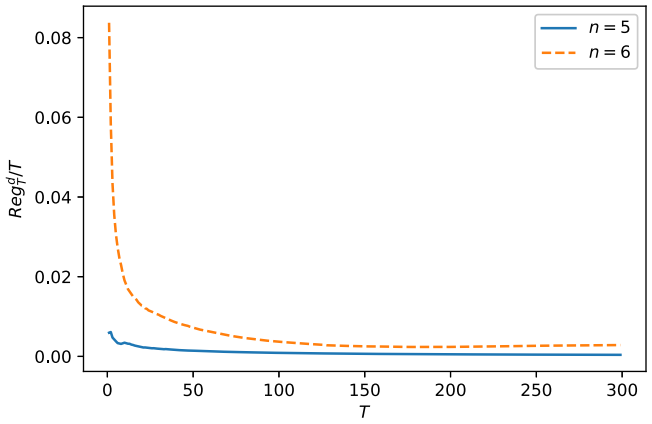


Fig. 5. Performances of proposed algorithm when n is different in the first experiment.

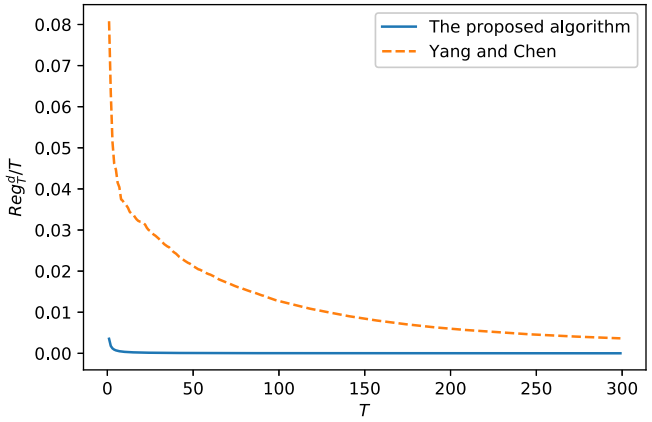


Fig. 6. Performances of proposed algorithm and Yang and Chen (2020) in the first experiment.

$$\begin{pmatrix} 0.1 & 0.3 & 0.2 & 0.2 & 0.2 & 0 & 0 \\ 0 & 0.5 & 0.4 & 0 & 0 & 0.1 & 0 \\ 0 & 0 & 0.5 & 0 & 0.4 & 0 & 0.1 \\ 0 & 0.5 & 0 & 0.5 & 0 & 0 & 0 \\ 0.25 & 0 & 0 & 0.25 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0.5 & 0 & 0.5 \end{pmatrix}, C = R^{\top}.$$ From Fig. 9, we know that when n is larger, the average dynamic regret is also larger, which corresponds to the result in Theorem 2.

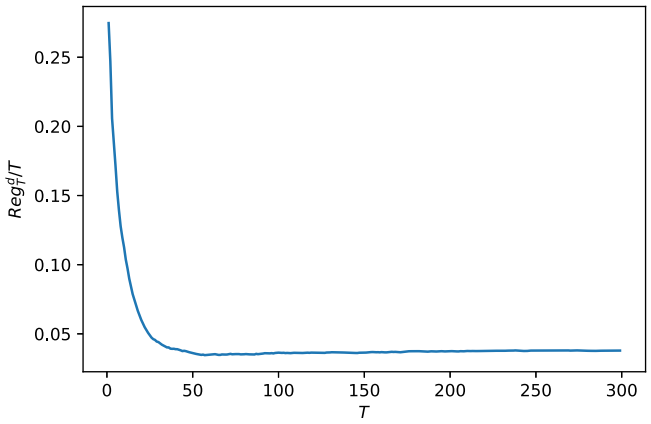


Fig. 7. Performance of proposed algorithm in the second experiment.

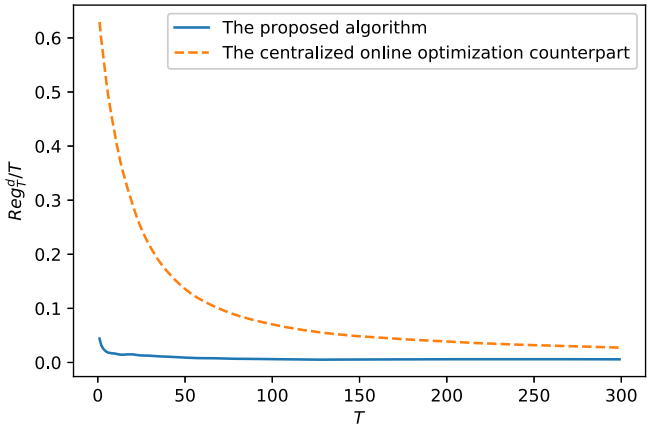


Fig. 8. Performances of proposed algorithm and its centralized counterpart in the second experiment.

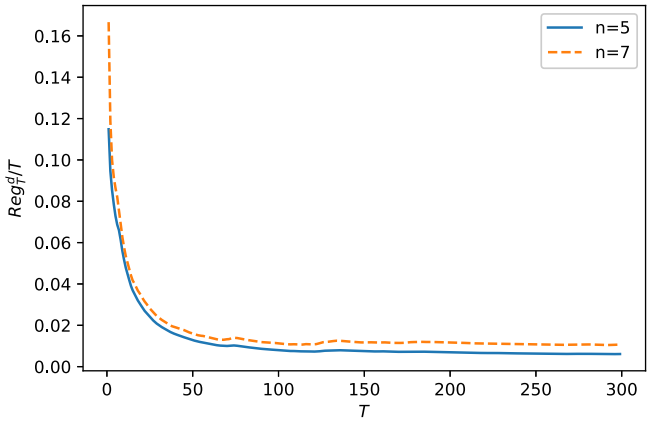


Fig. 9. Performances of proposed algorithm when n is different in the second experiment.

Fig. 10 shows that proposed algorithm outperforms Yang and Chen (2020) while setting our step size to be 0.025, while the step size of Yang and Chen (2020) is set to be 0.002.

6. Conclusion

In this paper, we proposed a novel online distributed optimization algorithm under unbalanced directed graphs, which is based on an offline Push–Pull algorithm. Also, we proved that

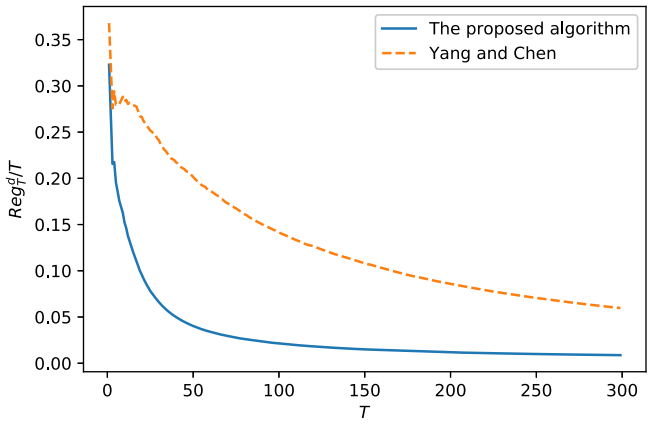


Fig. 10. Performances of proposed algorithm and Yang and Chen (2020) in the second experiment.

the dynamic regret of our proposed algorithm is sublinear when the path length and the gradient variance are also sublinear, and we used two numerical experiments to verify our theoretical results. Future work can be dedicated to tightening the upper bound of dynamic regret of online optimization algorithms; considering online distributed optimization problems with limited communication data rates, where we can apply quantization scheme to them; extending our research on Riemannian online distributed convex optimization, which has more applications in realistic; and thinking about online distributed optimization problem when loss function is nonconvex.

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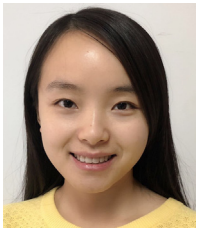
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